Assignment Overview: Structure Unstructured Restaurant Data with an LLM

Estimated time: 5 minutes

Introduction

This lab is the first step toward building an AI-powered multimodal restaurant recommendation system. Modern recommendation systems require more than numerical ratings. To generate meaningful recommendations, they rely on a rich, structured understanding derived from unstructured data.

In this lab, you will work with raw restaurant description text and apply **Large Language Models (LLMs)** to extract meaningful attributes and transform them into a structured, machine-readable format.

This lab focuses on the foundational step of GenAI-powered data pipelines: converting unstructured text into structured JSON suitable for search, validation, and downstream recommendation tasks.

Screenshot requirement for the Final Project

During the course of the lab, you will be instructed to take a screenshot of the Python code, clearly showing your implementation and the output for a particular task. Name the screenshot `M1L1_structure_for_loop.jpg`. This screenshot will later serve as a component for the Final Project.

Background

The structured JSON dataset generated in this lab forms the data foundation of the AI-powered multimodal **Restaurant Recommendation System**. It will be used to construct vector indexes for retrieval, support agent-based reasoning workflows, and consistency of this data directly influence the quality of recommendations and overall system performance.

Why this lab matters

You are learning how LLMs can be used as data transformation engines, not just text generators. By designing prompts, validating outputs, and repairing errors, you will gain hands-on experience building reliable, production-ready GenAI workflows that bridge unstructured data and structured knowledge bases.

What you will do in this lab

You will follow a step-by-step workflow to transform unstructured restaurant descriptions into a structured JSON knowledge file using LLMs. The sections that follow outline this workflow.

Explore unstructured restaurant data

To begin this process, you will load and inspect raw restaurant description text files. This step helps you understand the variability, noise, and semantic richness of real-world data, and identify key attributes such as **cuisine type**, **pricing**, **ratings**, and **signature dishes**.

Design prompts for structured extraction

Next, you will design a one-shot prompt template that instructs an LLM to extract predefined restaurant attributes and return them in a consistent JSON format. This step emphasizes clarity, structure, and output constraints to guide the model toward reliable data extraction.

Validate outputs and implement auto-repair

LLM outputs are not always perfectly formatted. Therefore, you will implement schema-based validation to detect malformed JSON. You will then introduce a specialized LLM-based repair mechanism that automatically corrects formatting errors, improving robustness and reliability.

Build a structured restaurant knowledge file

Finally, you will apply your pipeline across all restaurant descriptions using a loop to generate a complete JSON-formatted dataset. This structured file will serve as a reusable knowledge artifact for future labs and recommendation system development.

Deliverables

In this lab, you will complete the following deliverables:

- Design a one-shot prompt template to extract structured restaurant attributes from raw text
- Implement JSON schema validation to ensure output correctness
- Build an LLM-based auto-repair mechanism for malformed outputs
- Generate and save a fully structured JSON file containing all restaurant data

Conclusion

By the end of this lab, you will have transformed unstructured restaurant descriptions into a validated, structured knowledge base using LLMs. This workflow demonstrates how GenAI can be integrated into real-world data engineering pipelines to enable scalable, explainable, and intelligent applications.

Author(s)

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